



THE UNIVERSITY *of* EDINBURGH
School of Mathematics



Multiple imputation of missing transcriptomic data

Applications of Bayesian deep learning
ISBA World Meeting 2026 Short Course

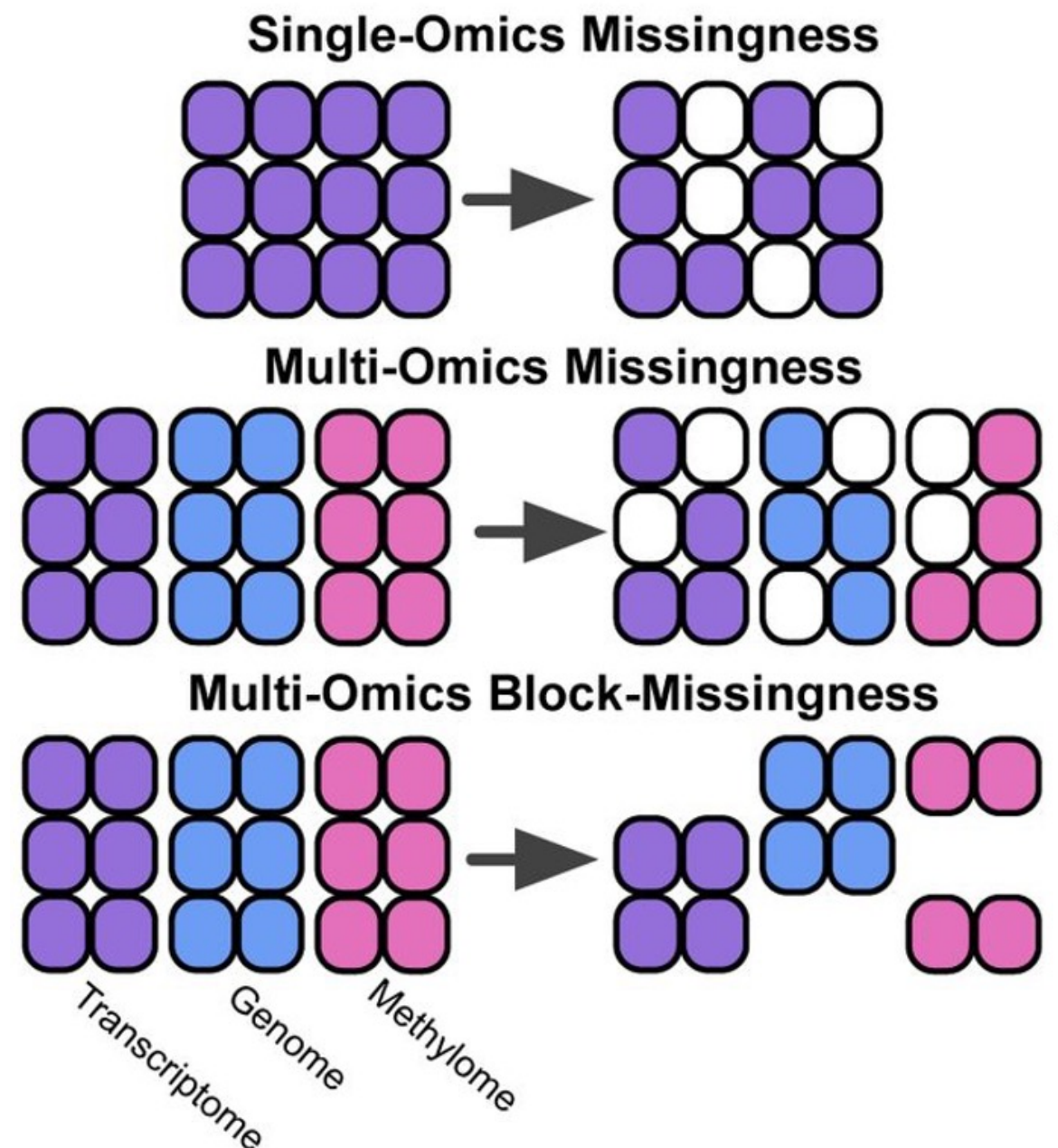
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Joint work with Breeshey Roskams-Hieter (Cambridge), Jude Wells (Imperial), Lucia Filippozzi (Trento), & Claudio Agostinelli (Trento)

Missing data

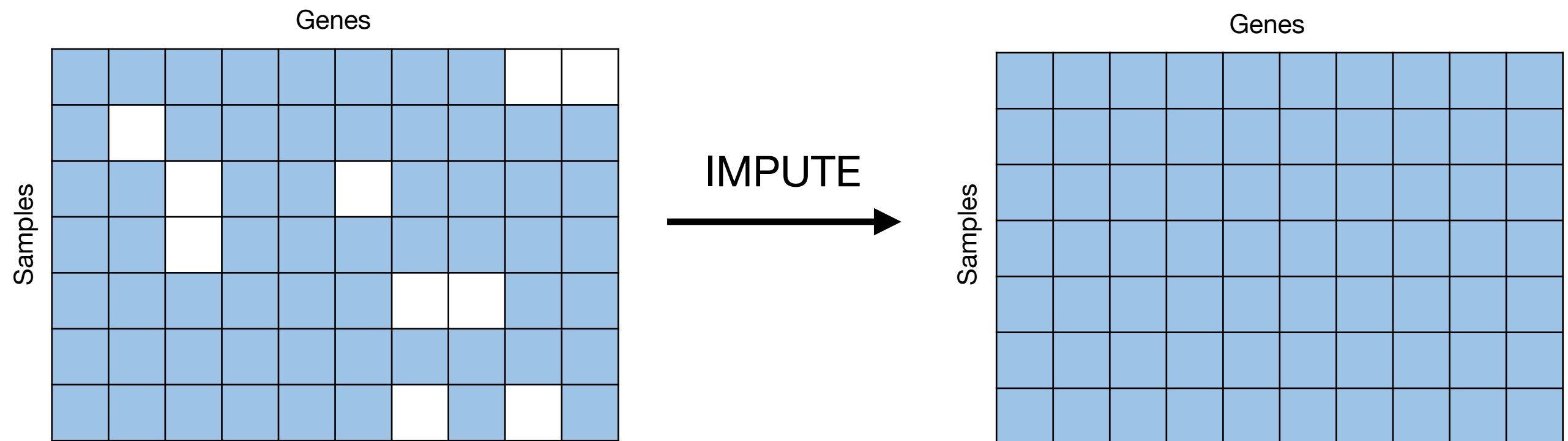
Examples in transcriptomics and multi-omics

- Missing data persists across many health data settings, yet full datasets are required for most downstream analyses
- Focus: transcriptomics, which suffer from inherent missingness due to the stochasticity of the biological processes and technical limitations.
- Recently, multi-omics exacerbate missingness issues, due to the necessity of combining multiple studies, often creating block-wise missingness.



Single imputation in transcriptomics

How do we handle missingness in high-dimensional omics data?



- Simple strategies (e.g. mean) may result in large imputation errors
- Recently, [deep learning](#) offers improvements, through its ability to capture complex relationships in high-dimensional omics data

What deep learning approaches can we use?

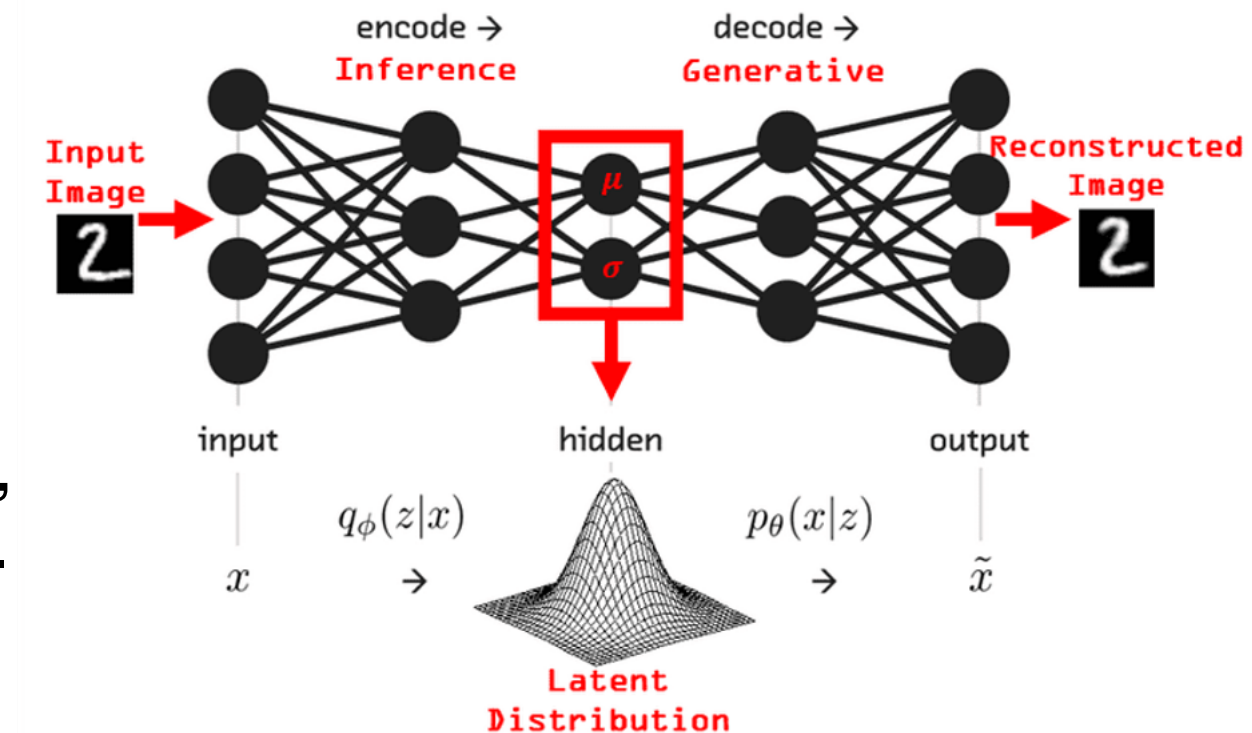
→ Variational autoencoders (VAEs)

Generative model: $p_{\theta}(x | z)$

- models the observed single-omic data $x \in \mathbb{R}^p$, given latent variables $z \in \mathbb{R}^d$ with $d \ll p$.
- Likelihood form (e.g. Gaussian, Poisson, negative binomial) depends on the data.
- Neural network (NN) maps z to the likelihood parameters (e.g. mean and variance for Gaussian), parametrised by θ .

Prior on the latent space: $\pi(z)$

- E.g. standard normal.

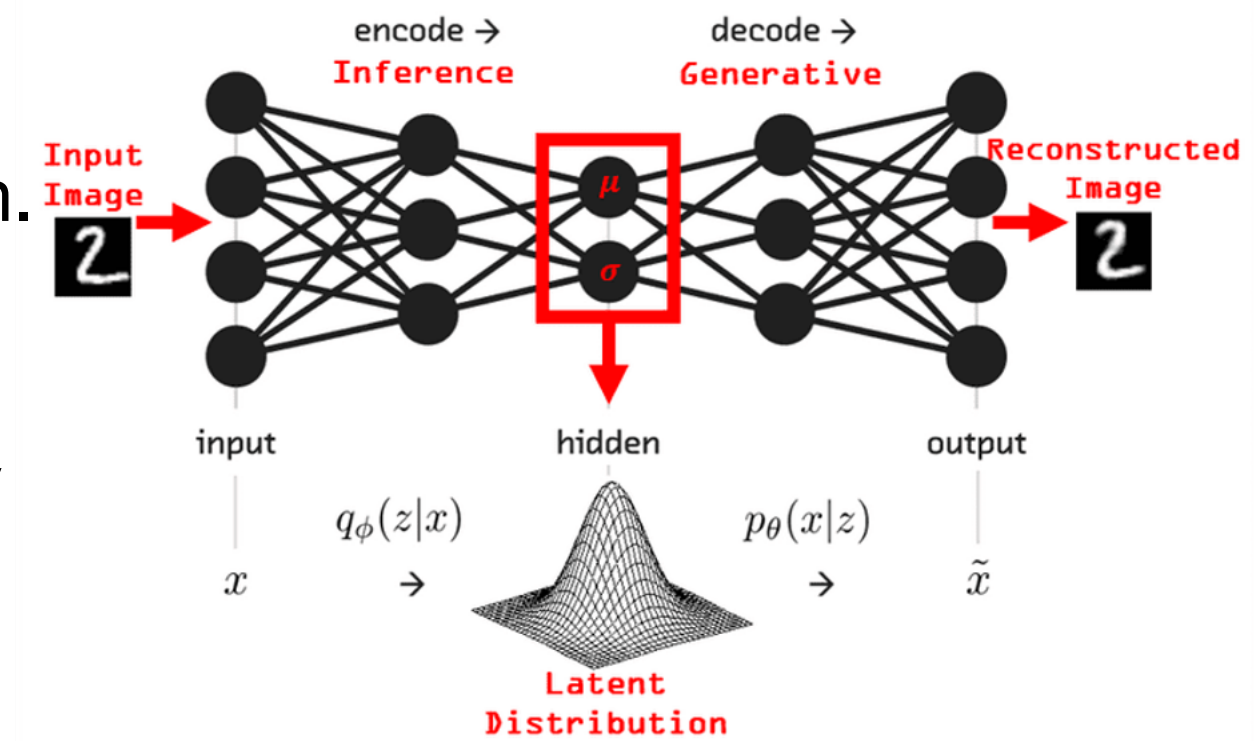


Variational autoencoders

How do we perform inference?

Variational posterior: $q_{\phi}(z | x)$

- Typically assumed factored Gaussian.
- Neural network (NN) maps x to the mean and variance of the variational Gaussian posterior, parametrised by ϕ .



Parameters θ and ϕ are found by maximising the ELBO:

$$\text{ELBO} = \sum_{i=1}^n \mathbb{E}_{q_{\phi}}[\log(p_{\theta}(x_i | z_i))] - \beta \sum_{i=1}^n \text{KL}(q_{\phi}(z_i | x_i) || \pi(z_i))$$

→ Variational inference for z + 'empirical Bayes' for θ

Variational auto-encoders

Why is there a parameter β ?

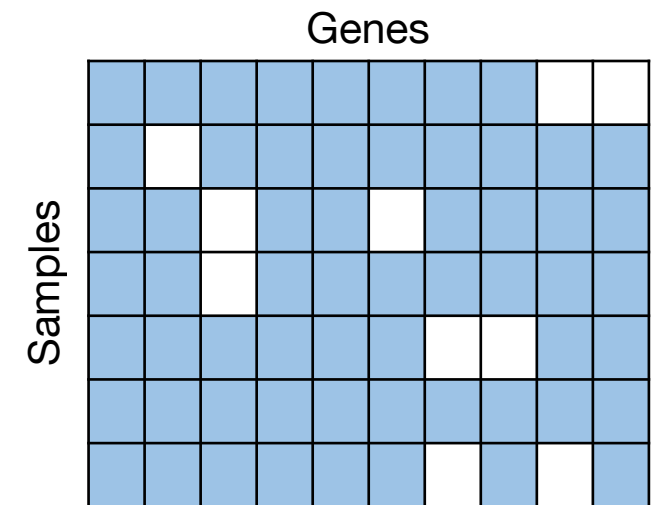
Parameters θ and ϕ are found by maximising the **ELBO**:

$$\begin{aligned} & \max_{\theta, \phi} \sum_{i=1}^n \mathbb{E}_{q_{\phi}}[\log(p_{\theta}(x_i | z_i))] - \beta \sum_{i=1}^n \text{KL}(q_{\phi}(z_i | x_i) || \pi(z_i)) \\ & \Leftrightarrow \max_{\theta, \phi} \sum_{i=1}^n \mathbb{E}_{q_{\phi}}[\log(p_{\theta}(x_i | z_i)^{1/\beta})] - \sum_{i=1}^n \text{KL}(q_{\phi}(z_i | x_i) || \pi(z_i)) \end{aligned}$$

- β controls the **strength of regularisation**
- Equivalent to using the **power likelihood** with power $\lambda = 1/\beta$

Imputation with VAEs

How to train VAEs and impute in the presence of missing data?

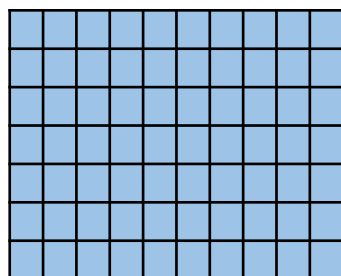


First, train a VAE (on complete subset or mean imputed data).

Single imputation: let $\hat{x}_i = (\hat{x}_i^{\text{mis}}, x_i^{\text{obs}})$

- Set $\hat{z}_i = \mathbb{E}_{q_\phi}[z_i | \hat{x}_i]$
- Set $\hat{x}_i^{\text{mis}} = \mathbb{E}_{p_\theta}[x_i^{\text{mis}} | x_i^{\text{obs}}, \hat{z}_i]$

→ Iterate until convergence

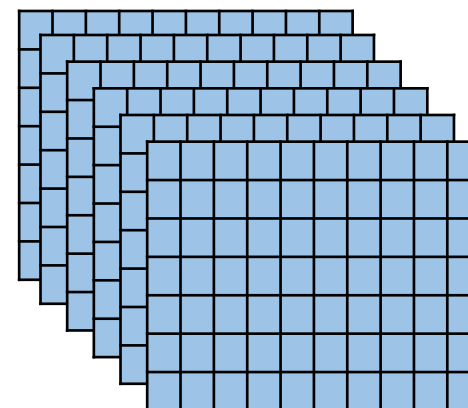


Multiple imputation: (e.g. importance sampling)

- Sample $z_i^{(s)} \sim q_\phi(\cdot | \hat{x}_i)$, $s = 1, \dots, S$
- Compute importance weights

$$w_i^{(s)} \propto \frac{p_\theta(x_i^{\text{obs}} | z_i^{(s)})^{1/\beta} \pi(z_i^{(s)})}{q_\phi(z_i^{(s)} | \hat{x}_i)}, \quad s = 1, \dots, S$$

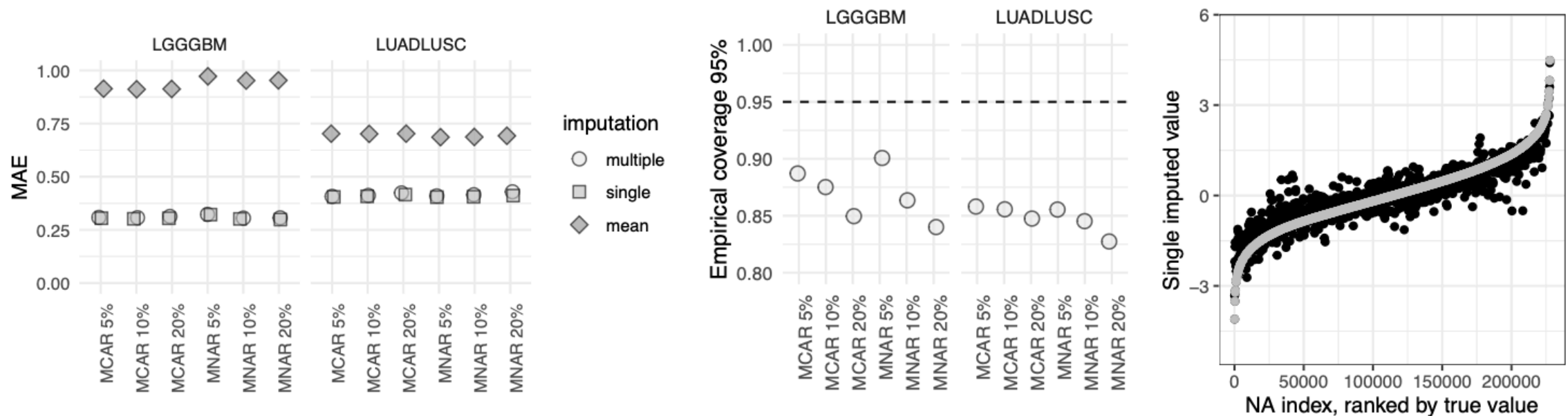
- Resample $z_i^{(s)}$ with weights $w_i^{(s)}$
- Sample $x_i^{\text{mis}(s)} \sim p_\theta(\cdot | x_i^{\text{obs}}, z_i^{(s)})^{1/\beta}$



Multiple datasets reflecting uncertainty in missing values

How well do standard VAEs impute missing data?

Good accuracy, but **overconfident** imputations, esp. for more extreme values



(a) Comparison of MAE

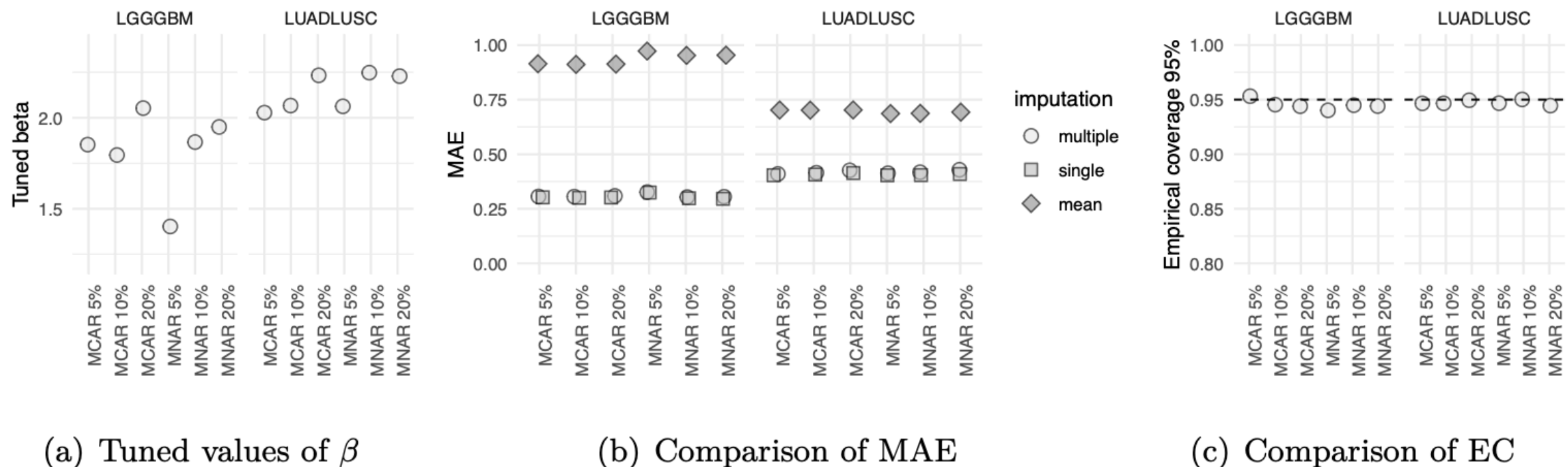
(b) Comparison of EC

(c) Single imputed values:
LGG/GBM, MCAR 10%

- Two transcriptomic datasets from the Cancer Genome Atlas (TCGS), recording gene expression for $p = 17,175$ genes.
- The glioblastoma and lower-grade glioblastoma (LGGGBM) dataset contains $n = 667$ patient samples; the lung adenocarcinoma and lung squamous carcinoma (LUADLUSC) contains $n = 953$ patient samples.
- Missing values generated by randomly masking (MCAR) and by randomly masking those with high GC content (MNAR) to mimic realistic dropout during sequencing.

Do β -VAEs improve uncertainty in imputations?

Appropriately tuned β provides nominal coverage

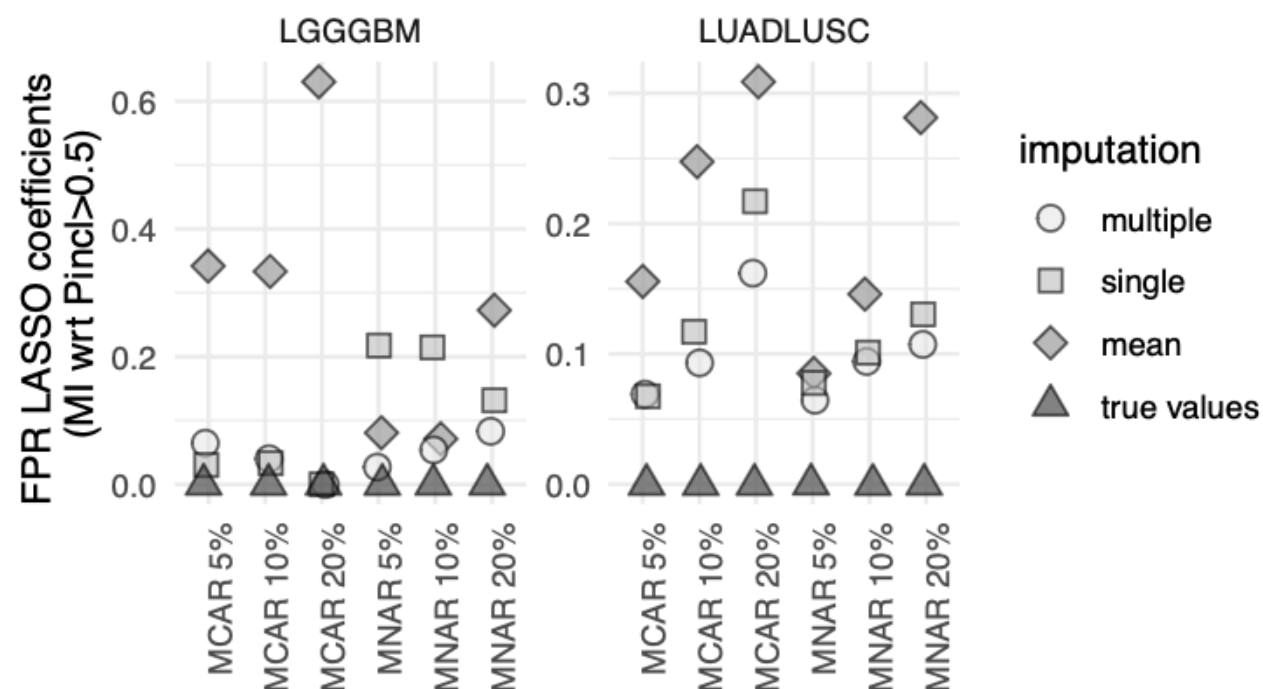


- Selecting β is important; and when appropriately tuned, β -VAEs provide **improved coverage**, yet similar accuracy (tuning strategies in Filippozzi et al (2026)).

What is the impact on downstream analyses?

Identifying genes associated with cancer types

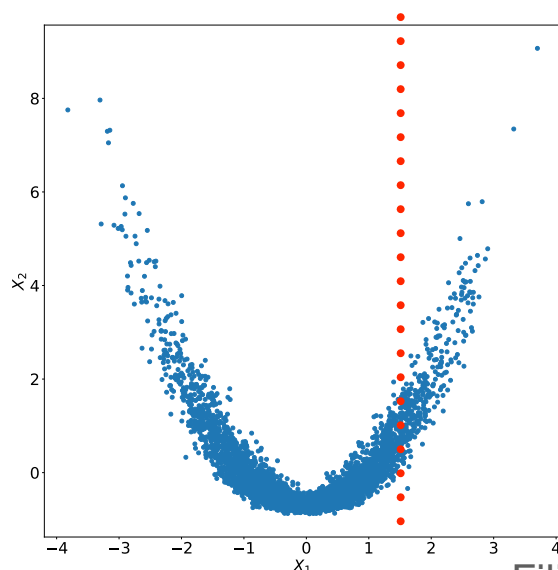
- Discriminating gene sets are identified via logistic regression with L_1 regularisation and compared with the set identified using the complete data.
- In multiple imputation, empirical probabilities of inclusion across the multiple fits (datasets) are used to identify gene sets.
- Multiple imputation with tuned β - VAEs **reduces false discovery** in downstream analysis.



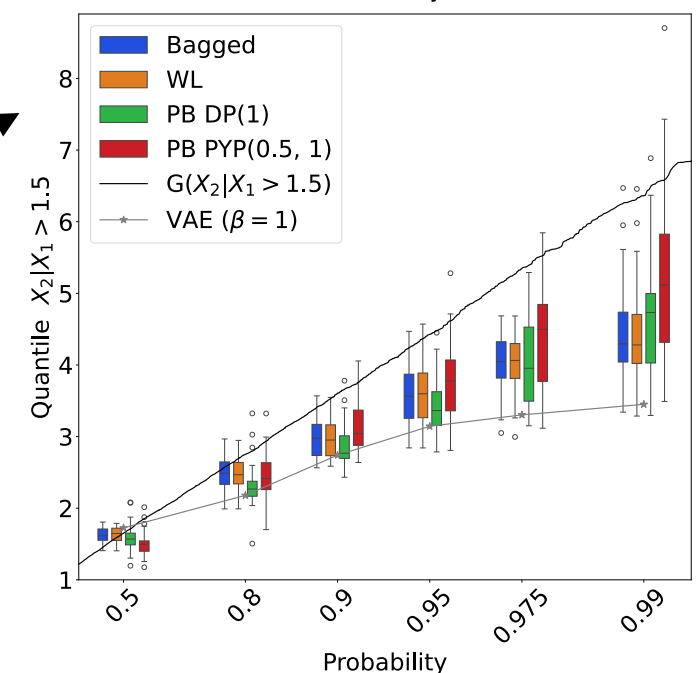
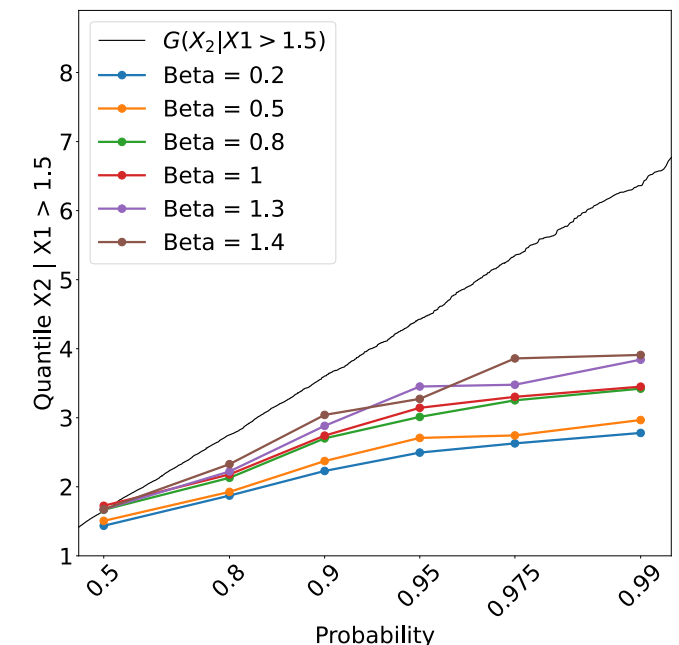
Can we take a more Bayesian approach?

Yes! Uncertainty in the NN parameters → **robust imputation**

- β -VAEs rely on point estimates of the NN parameters and requires both i) careful tuning of β and ii) initial imputation or row-wise deletion.
- Bootstrap-based approaches can overcome these limitations, providing:
 - i. computationally efficient approximation of the posterior,
 - ii. uncertainty quantification,
 - iii. robustness to model misspecification,
 - iv. reduced variance of unstable estimators through bagging.



Compare and develop **bootstrap-based VAEs** for approximate Bayesian inference in deep latent variable models



Filippozi, W., Agostinelli (2026+) *Bootstrapping Variational Autoencoders*.

Newton et al (2021) *Weighted bayesian bootstrap for scalable posterior distributions*.

Fong et al (2019) *Scalable nonparametric sampling from multi-modal posteriors with the posterior bootstrap*.

Fan et al (2025) *Variational Bagging*.

Discussion

- Standard deep learning approaches (e.g. VAEs) can cause overfitting in single imputation → effects downstream tasks
- Multiple imputation through *Bayesian* deep learning retains imputation accuracy while measuring uncertainty around missing values
- Improves downstream analysis (e.g. less false positives)
- More generally, deep learning applications in biomedicine are growing → Bayesian approaches can help improve robustness in downstream analyses and decision-making

